

Annex III — Proof of Authoritative Learning (PoAL™)

Governed Evolution of the WR Graph™

Document status: Normative specification. This Annex is self-contained and carries no cross-references to other annexes.

Applies to: WR Desk™ runtime, WR Graph™, Workflow Context Recorder (WCR), BEAP™-adjacent learning operations, and all optimization processes executed during idle periods ("Night Pipeline").

Trademark notice: Optirando™, WR Desk™, WR Graph™, BEAP™, PoAE™, PoAL™, and WR Code® are used throughout as product and protocol designations of the Optirando™ system.

§1 Purpose and Scope

§1.1 Motivation

The WR Graph™ is defined elsewhere in the Optirando™ Canon as the unifying context layer of the system, combining four knowledge dimensions: **declarative** (facts, documents, retrieval sources), **episodic** (execution history, feedback events, outcomes), **procedural** (skills, WCR recordings, workflows, prompt profiles), and **parametric** (adapters and other trained artifacts).

The Canon specifies what the WR Graph™ contains. This Annex specifies **how the WR Graph™ is permitted to change**. It is the dynamic complement to the static definition: no dimension of the WR Graph™ holds a right of mutation without demonstrable authority.

§1.2 Core Principle

Proof of Authoritative Learning (PoAL™) extends the trust model of Proof of Authoritative Execution (PoAE™) from the execution boundary to the learning boundary:

Where PoAE™ guarantees that no action is executed without an authorized, auditable path, PoAL™ guarantees that no knowledge is learned, no capability is acquired, and no graph state is mutated without an authorized, auditable path.

The system's optimization objective is **Long-Term Local Capability Growth**: project knowledge and automation capability are improved iteratively across all four knowledge dimensions, using idle compute time, on local hardware, under local control.

§1.3 What PoAL™ Governs

PoAL™ governs every mutation of the WR Graph™, including but not limited to:

1. Ingestion of new declarative sources (RAG corpus changes)

2. Creation and revision of procedural entities (skills, WCR recordings, prompt profiles, validators, workflows)
3. Creation, training, activation, and retirement of parametric artifacts (adapters, specialized models)
4. Derivation and condensation of episodic knowledge (feedback events, outcome aggregation)
5. Routing and confidence metadata affecting any of the above
6. Any interaction with cloud services in service of learning

§1.4 What PoAL™ Does Not Govern

PoAL™ does not govern task execution at runtime (governed by PoAE™), transport security (governed by the sealed transport specification), or vault semantics (governed by the vault specification). PoAL™ does not impose any legal interpretation on the operator; it records provenance and enforces operator-defined policy (see §3.6).

§2 Definitions

Mutation. Any create, update, invalidate, or delete operation on a WR Graph™ entity or edge.

Evidence Event. A persisted, timestamped record of an observable signal from daytime operation: a user feedback action, a run outcome, a retrieval-quality measurement, or a correction.

Gap. A typed, evidence-backed hypothesis that the system's current knowledge or capability is insufficient for a recurring class of work. A gap without linked evidence is invalid and MUST NOT be surfaced.

Proposal. A concrete, reviewable remediation for a gap or an improvement opportunity, produced by the Night Pipeline and presented in the Morning Digest.

Consent Record. A persisted, timestamped record of an operator decision (accept / reject / defer) on a proposal, bound to the proposal's identity and content hash.

Morning Digest. The bounded set of proposals presented to the operator after a Night Pipeline run.

Night Pipeline. The ordered sequence of idle-time optimization stages defined in §6.

Provenance Chain. The complete derivation record of a graph entity: its originating evidence, its consent record(s), its source(s), and its downstream dependents.

§3 Normative Area 1 — Cross-Dimensional Provenance Model

§3.1 Universal Provenance Requirement

Every entity in the WR Graph™, in every knowledge dimension, MUST be able to answer three questions at any time:

1. **Origin** — From which evidence event(s), source(s), and consent record(s) does this entity derive?

2. **Derivation** — Which upstream entities does it build upon (data sets, sources, prior versions)?
3. **Dependency** — Which downstream entities depend on it (the deletion cascade)?

An entity for which any of the three questions is unanswerable is **provenance-orphaned** and MUST be flagged in the next Morning Digest.

§3.2 Rationale (Informative)

Provenance is required for reasons independent of any legal consideration:

- **Regenerability.** Parametric artifacts are bound to a base model. Base models change. The durable asset is the data set and the training recipe, not the weights. Regeneration on a new base model requires the full derivation chain.
- **Diagnosis.** When an artifact underperforms, the first diagnostic question is which source introduced the defect.
- **Deletion cascade.** When a source is withdrawn (contract end, operator decision, data-subject request), every derived entity must be identifiable and re-derivable or retirable.
- **Auditability.** Organizational deployment requires that any capability of the system can be traced to an authorized acquisition path.

§3.3 Source Entities

Every external source (document, book, manual, API documentation, web resource, purchased material, cloud-generated content) MUST be represented as a Source entity with at minimum:

Field	Requirement
source_id	Stable identifier
acquisition	How obtained (user-provided / discovered / purchased / cloud-generated / WCR-derived)
acquired_at	Timestamp
consent_ref	Consent record authorizing ingestion
license_status	cleared / unclear / restricted — as declared by the operator
training_permitted	yes / no / unclear — as declared by the operator
content_hash	Integrity binding

The system records these fields; it does not adjudicate them. Declaration is an operator act.

§3.4 Data Set Entities

Training data sets are first-class graph entities. A data set MUST reference all Source entities it draws from, the generation method (extraction / synthesis / paraphrase / WCR-derivation), the generating component (local model / cloud service, with the cloud abstraction boundary of §7 applying), and quality metrics from its evaluation runs.

Data sets, not adapters, are the persistent asset. Adapters are treated as compiled caches of data sets against a specific base model and are regenerable by design.

§3.5 Deletion Cascade

When a Source entity is invalidated, the system **MUST** identify all dependent data sets, adapters, prompt profiles, and RAG index segments, and **MUST** present a cascade plan (retire / regenerate without the source / retain with flag) as a Class-3 proposal (§4). Silent retention of derivatives of an invalidated source is prohibited.

§3.6 Policy Gate (Configurable)

Whether entities with `license_status: unclear` or `training_permitted: unclear` may enter the training path is a **policy configuration**, not a fixed system rule:

- **Default (single-operator deployments):** open — provenance is recorded, no gate is enforced.
- **Organizational policy:** a deployment **MAY** enforce that only `cleared/yes` sources enter data-set generation and training.

The policy setting itself, and every change to it, is a logged graph mutation.

§3.7 Temporal Validity

All provenance-bearing edges carry bi-temporal validity (`valid_from`, `valid_until`, `recorded_at`). Contradicting knowledge invalidates prior edges rather than deleting them. Confidence values decay in the absence of confirming evidence; decay parameters are deployment configuration.

§4 Normative Area 2 — Mutation Class Taxonomy

§4.1 Principle

Learning is continuous and fine-grained; consent is a scarce operator resource. Requiring consent for every mutation would render the system inoperable; requiring consent for none would render it ungoverned. PoAL™ therefore partitions all mutations into three classes. **Every mutation operation implemented in the system **MUST** be assigned to exactly one class**, and the assignment table (§4.5) is normative.

§4.2 Class 1 — Autonomous Hygiene

Mutations that are purely derived, reconstructible from existing data, and do not introduce new knowledge or capability.

- Executed autonomously during the Night Pipeline.
- Logged (append-only), not surfaced in the Morning Digest except in aggregate ("hygiene summary").
- **MUST** be reversible by recomputation.

Examples: confidence updates and decay, embedding re-indexing over already-ingested content, linking evidence events to existing entities, episodic condensation that preserves the underlying events, statistics aggregation.

§4.3 Class 2 — Reported Adjustments

Mutations that influence system behavior but are trivially reversible and carry no acquisition of new knowledge, no new capability, and no external contact.

- Executed autonomously, **reported item-by-item** in the Morning Digest with a one-action rollback.
- MUST retain the prior state for rollback for a configurable retention window.

Examples: routing weight adjustments among existing implementations, retrieval parameter tuning within an existing corpus, re-ranking of existing skills for an existing context, activation-threshold adjustments.

A deployment MAY, by policy, promote any Class-2 operation to Class 3.

§4.4 Class 3 — Consent-Gated Mutations

Mutations that introduce new knowledge or new capability into the WR Graph™, alter procedural behavior, or involve any external contact.

- MUST NOT be applied without an explicit Consent Record.
- Presented as proposals in the Morning Digest, each carrying its evidence chain (§5), its provenance plan (§3), and where applicable a before/after demonstration.
- Rejected proposals are persisted with their rejection; rejection dampens future proposals of the same type (§5.6).

Examples: ingestion of any new source; creation or revision of prompt profiles, validators, WCR recordings, or workflows; data-set generation; adapter training, activation, or retirement; deletion cascades; any cloud interaction in service of learning; policy changes.

No Class-3 mutation is ever applied silently. This mirrors, and is normatively identical in spirit to, the WCR self-optimization rule: propose, never apply silently.

§4.5 Assignment Table (Normative)

Operation	Class
Confidence update / decay	1
Embedding re-index (existing corpus)	1
Evidence-event linking	1
Episodic condensation (events preserved)	1
Recurring-action pattern detection (opt-in, §5.7)	1
Routing weight adjustment	2
Retrieval parameter tuning (existing corpus)	2
Skill re-ranking for context	2
New RAG source ingestion	3
Prompt profile creation / revision	3
Validator creation / revision	3
WCR recording revision	3
Data-set generation	3
Adapter training / activation / retirement	3

Operation	Class
Deletion cascade execution	3
Any learning-related cloud interaction	3
Policy configuration change	3

Implementations **MUST** extend this table before introducing any new mutation operation; an unclassified mutation operation is a specification violation.

§5 Normative Area 3 — Evidence, Feedback Events, and Gap Typology

§5.1 Evidence Requirement

A gap is valid only if it is backed by linked evidence events. The Night Pipeline **MUST NOT** surface any gap or proposal that cannot present its evidence. "Topic occurred" is not evidence of a gap; "topic recurred **and** performance was demonstrably insufficient" is.

§5.2 Feedback Event Schema

The primary evidence signal is explicit operator feedback. A Feedback Event **MUST** persist:

Field	Content
event_id	Stable identifier
target_ref	The output, run, or WCR execution being rated
skill_ref	The procedural entity that produced the target (if any)
verdict	positive / negative
commentary	Operator free text (optional but strongly weighted)
project_ref	Project / domain context
timestamp	Event time

Secondary evidence signals (run outcomes, retrieval-quality scores, reformulation detection, correction diffs) follow the same event pattern and **MAY** be added incrementally; their absence does not block V1 operation (§6.6).

§5.3 Baseline Check

Before a knowledge gap may be declared, the Night Pipeline **MUST** verify that the deficiency is real: the affected queries of the day are re-evaluated against the base model plus the existing corpus. A gap is declared only where the baseline demonstrably fails. Commodity knowledge already mastered by the base model **MUST NOT** generate proposals.

§5.4 Gap Typology (Normative)

Every Gap entity carries exactly one type; the type determines the remediation path and its cost class:

Gap type	Meaning	Remediation path (ascending cost)
Knowledge gap (external)	Required knowledge exists publicly but not in the corpus	Source discovery → RAG ingestion → prompt profile → (last) data set + adapter

Gap type	Meaning	Remediation path (ascending cost)
Knowledge gap (internal)	Required knowledge exists only with the operator/organization	Elicitation: the system requests the knowledge from the operator (dictation, document, WCR recording)
Capability gap	Knowledge suffices; a tool or renderer is missing	Tool integration → workflow
Process gap	A recurring procedure has no recorded skill	WCR recording proposal → workflow compilation
Quality gap	Outputs are produced but repeatedly corrected	Validator → prompt profile revision → (last) adapter

§5.5 Internal Knowledge Elicitation

The internal knowledge gap is a first-class mechanism, not a fallback. Where evidence shows recurring reference to undocumented internal matter (processes, decisions, conventions), the Night Pipeline SHALL generate an elicitation proposal ("Process Y is referenced repeatedly but described nowhere — record it?"). Elicited knowledge enters the graph as a Source with **acquisition: user-provided** and full provenance.

§5.6 Digest Budget and Meta-Feedback

- The Morning Digest is **hard-bounded** (default: five proposals), ranked by expected utility × evidence strength.
- Every proposal outcome (accepted / rejected / ignored) is itself an evidence event on the proposing mechanism. Rejected proposal types **MUST** dampen future ranking of that type. The operator trains the proposer.

§5.7 Recurring-Action Detection (Opt-In)

Where enabled by configuration, the system detects **recurring action patterns** in episodic evidence — sequences of manual operations, queries, or executions that repeat across sessions within the same semantic context — and treats them as candidate process gaps (§5.4).

- Detection itself operates exclusively on locally recorded episodic events and is a Class-1 operation (derived, reconstructible, no new knowledge).
 - Where a pattern exceeds a configurable recurrence threshold, the Night Pipeline SHALL generate a **WCR recording proposal** for the Morning Digest ("This sequence has recurred N times in project Z — record it as a reusable automation?"), carrying the recurrence evidence as its evidence chain.
 - The proposal is Class 3: no recording session is initiated, and no procedural entity is created, without an explicit Consent Record.
 - An accepted proposal leads to an operator-performed WCR recording; the resulting skill enters the graph with **acquisition: user-provided** provenance and is routed per §6.4.
 - The feature is **opt-in**: with detection disabled, no recurrence analysis over episodic events is performed.
-

§6 Normative Area 4 — Night Pipeline and Escalation Ladder

§6.1 Pipeline Stages (Ordered)

The Night Pipeline executes in fixed order; each stage is independently useful and independently interruptible:

1. **Graph hygiene** (Class 1): confidence decay, re-indexing, event linking, episodic condensation.
2. **Baseline re-evaluation** (Class 1): the day's affected queries evaluated against base model + existing corpus; parametric artifacts A/B-checked against the bare base model on real queries of the day.
3. **Gap analysis** (produces Class-3 proposals): typed gaps with evidence chains, including recurring-action patterns where detection is enabled (§5.7), ranked into the Digest budget.
4. **Remediation drafting** (produces Class-3 proposals): concrete artifacts — revised prompt profiles counter-evaluated against the objected examples ("would the revision have addressed the critique?"), source-discovery results, validator drafts, elicitation requests.
5. **Skill maintenance** (Classes 1–3 as applicable): WCR endpoint indexing, precondition edges (§6.4), drift detection.
6. **Data-set generation and training** (Class 3, consent required per step): executed only for accepted gaps that the preceding stages demonstrably could not close, and only from Sources admitted by the active policy gate (§3.6). Training uses the locally configured fine-tuning stack; resulting adapters enter the graph with full provenance and telemetry hooks (§6.3).

§6.2 Escalation Ladder

Remediation cost ascends; the Pipeline MUST exhaust cheaper rungs before proposing more expensive ones:

Workflow compilation → prompt profile → RAG enrichment → validator → tool integration → adapter training.

Adapter training is justified only where (a) a gap is persistent across baseline checks, (b) cheaper rungs have been tried or ruled out with recorded rationale, and (c) sufficient admissible data exists. The expected steady state is that most gaps close below the adapter rung.

§6.3 Parametric Telemetry and Retirement

Every active adapter MUST accumulate per-adapter telemetry: acceptance rate versus correction rate, and periodic idle-time A/B comparison against the bare base model. An adapter that no longer outperforms its base model — including after a base-model upgrade — MUST be proposed for retirement (Class 3). Retired adapters remain in the graph as provenance records; their data sets remain available for regeneration.

§6.4 Skill Routing (Two-Stage)

Procedural skills (including WCR endpoints stored directly in the graph) are routed in two stages:

1. **Semantic stage:** embedding retrieval over skill descriptions yields candidates for the current context.
2. **Precondition stage:** graph traversal filters candidates by recorded preconditions — target system reachability, project context, UI/version compatibility, authorization.

Below a confidence threshold, a matching skill is **offered**, not executed; above it, it may enter the standard execution preview. Routing is a graph property (edges: skill → precondition, skill → project, skill → confidence), not a separate subsystem.

§6.5 Temporal Complement to Intent Validation (Informative)

Daytime intent validation asks whether the question posed is the right question. The Night Pipeline asks the temporal complement: whether the knowledge exists to pose the right question at all. Gap analysis is intent validation extended into the time dimension.

§6.6 Minimal Viable Loop (V1)

A conforming minimal implementation consists of: the Feedback Event schema (§5.2), Class-3 proposal generation restricted to **prompt profiles and RAG suggestions**, the Digest with budget and meta-feedback (§5.6), and Class-1 hygiene. Baseline checks, elicitation, source discovery, telemetry, and training are conforming extensions in that order. No stage of V1 requires training, cloud access, or any license consideration.

§7 Normative Area 5 — Cloud Abstraction Boundary

§7.1 Principle

Cloud services MAY assist learning; they MUST NOT receive operational knowledge. The boundary is defined as a **data-class allowlist**, not as a list of permitted functions, because functions leak differently depending on prompt content.

§7.2 Permitted Outbound Data Classes

Only the following MAY cross the boundary, and only under a Class-3 consent for the interaction:

1. **Taxonomy-level skill descriptors** — generic subject identifiers ("OAuth token refresh", "invoice validation", "polynomial simplification") with no instance data.
2. **Generic generation requests** — requests for synthetic examples, benchmark templates, or curriculum structures phrased exclusively at taxonomy level.
3. **Public-literature queries** — searches over public material.

§7.3 Prohibited Outbound Data Classes

The following MUST NOT cross the boundary under any configuration:

- WR Graph™ structure, content, or excerpts, in any dimension
- WCR recordings or fragments thereof
- Evidence events, feedback commentary, or correction content
- Usage frequencies and recurrence statistics (metadata leaks business activity)

- Internal source content or identifiers
- Any confidential material as classified by the local classifier

§7.4 Local Instantiation Step

Cloud-generated artifacts are generic by construction. Mapping generic artifacts onto concrete projects — real field names, real processes, real documents — is a **mandatory local step** performed by local models only. Cloud output enters the graph as a Source with **acquisition: cloud-generated** and is subject to the same provenance and policy rules as any other source.

§7.5 Local-Only Guarantee

Every learning function specified in this Annex **MUST** be operable with cloud access disabled entirely. Cloud assistance is an accelerant, never a dependency. Degradation to local-only mode is a configuration act, not an architectural change.

§8 Relation to Prior Work (Informative)

This Annex composes known mechanisms under a governance model; the individual mechanisms are not claimed as novel.

- **Idle-time optimization:** sleep-time compute approaches (Letta, 2025) established that agent systems benefit from offline reflection between interactions. PoAL™ differs in placing the entire offline process under consent, evidence, and provenance obligations.
- **Skill accumulation:** skill libraries in embodied and workflow agents (Voyager, Wang et al., 2023; Agent Workflow Memory, 2024) demonstrated persistent capability growth. PoAL™ adds precondition-gated routing and provenance-bound lifecycle management.
- **Offline prompt optimization:** programmatic prompt optimization (DSPy / MIPROv2) is the pattern instantiated at the prompt-profile rung of the escalation ladder.
- **Adapter libraries and routing:** modular LoRA libraries with learned routing (PHATGOOSE and successors) are the nearest relatives of the parametric dimension; PoAL™ treats adapters as regenerable caches of provenance-bound data sets rather than as primary assets.
- **Active learning:** evidence-driven gap detection descends from active-learning practice; the internal-knowledge elicitation mechanism (§5.5) inverts conventional retrieval augmentation by identifying absent knowledge and requesting it from its only holder.

The claimed contribution is the normative composition: a knowledge graph in a governed system holds no right of mutation — in any of its dimensions — without demonstrable, auditable authority, and the learning process itself is subject to the same proof discipline as execution.

§9 Version History

Version	Date	Change
1.0	2026-07-07	Initial normative specification: cross-dimensional provenance model, mutation class taxonomy, evidence and gap typology, night pipeline and escalation ladder,

Version Date

Change

cloud abstraction boundary.

End of Annex III.